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VORTEX-HRM: HIERARCHICAL RETRIEVAL-AUGMENTED GENERATION FOR MULTI-HOP QUESTION ANSWERING

ABSTRACT. We present VORTEX-HRM, an agentic retrieval-augmented generation system for multi-hop question answering. Unlike prior work that relies on LLM-driven tool selection or flat ReAct loops, VORTEX separates planning from execution using a fact-free planner and a fixed chained retrieval pipeline. The Gravitational Core (planner) holds zero factual knowledge in its weights—it is a pure structural router communicating via an XML-based protocol. The Centrifugal Ingestor (executor) resolves sub-questions through a deterministic pipeline: keyword search, chunk reading, and LLM condensation. The cycle repeats—each rotation is a spiral—until convergence via entropic collapse. On a 50-question multi-domain benchmark (qwen2.5:7b CPU, 60 chunks, two runs), VORTEX achieves 70–72% Contains accuracy with 1.3 average spirals per question. EM is 0% due to 7B model output formatting limitations, not architecture. The system is offline-first (zero API keys), config-driven, and validated across three backends (mock, Ollama, OpenAI).

§1. INTRODUCTION

Multi-hop question answering requires retrieving and reasoning over multiple documents. Standard RAG [?] retrieves once and answers in a single step—it cannot chain evidence across documents.

Recent agentic approaches attempt to address this. A-RAG [?] augments RAG with LLM-driven tool selection, but this wastes tokens on routing decisions and introduces failure modes. ReAct [?] interleaves reasoning and acting but lacks a convergence signal—it terminates by hard step limit or the LLM’s arbitrary decision. RAPTOR [?] builds hierarchical document trees via recursive summarization, requiring expensive index-time preprocessing.

We introduce VORTEX, a system that replaces LLM tool selection with a fixed chained pipeline, replaces arbitrary termination with entropic collapse, and replaces parametric knowledge with fact-free routing. The system is offline-first, config-driven, and validated on CPU hardware.

Key words and phrases: retrieval-augmented generation and multi-hop QA and agentic systems and hierarchical retrieval.

§2. RELATED WORK

Retrieval-Augmented Generation. Standard RAG [?] combines a retriever with a generator. Self-Ask [?] decomposes questions via prompting. Chain-of-Thought [?] uses stepwise reasoning. VORTEX externalizes these steps into explicit retrieval cycles with structured state.

Agentic RAG. ReAct [?] alternates reasoning traces with tool calls. A-RAG [?] provides hierarchical retrieval interfaces for LLM-driven tool selection. GRASP [?] uses graph-based sub-agents with token-efficiency objectives. HERA [?] evolves multi-agent orchestration via an experience library. VORTEX differs from all of these by using a fixed chained pipeline (no LLM tool selection), a fact-free planner (zero parametric knowledge), and entropic convergence detection.

Hierarchical Retrieval. RAPTOR [?] builds tree-structured indices via recursive summarization. VORTEX uses flat chunks with iterative spirals—no index-time overhead.

§3. VORTEX ARCHITECTURE

3.1. Design Principles. VORTEX rests on four principles: (1) **Fact-Free Synapses**—the planner uses 0% of its capacity for facts and 100% for routing; (2) **Entropic Collapse**—convergence is detected via entropy plateau; (3) **Offline-First**—the default mock mode requires zero dependencies; (4) **Same Code, Any Hardware**—a config file switches between local CPU (Ollama) and cloud GPU (OpenAI).

3.2. Gravitational Core (Planner). The planner holds zero factual knowledge in its weights. Its state consists of: `goal_vector` (immutable original question), `spiral_memory` (accumulated `<fact>` entries), `hop_history` (full reasoning log), and confidence/entropy scores tracking convergence. It communicates via an XML contract: `<step>` emits a sub-question, `<final_answer>` emits the answer, and `<stop_search>` signals termination.

3.3. Centrifugal Ingestor (Executor). Unlike agentic systems where the LLM selects tools, VORTEX uses a fixed chained pipeline:

- (1) `keyword_search(query)`—lexical match returning top-k chunk IDs
- (2) `chunk_read(id, adjacent=True)`—full text with neighbor chunks
- (3) LLM condensation—compresses retrieved text into a structured `<fact>`

This is cheaper, more predictable, and equally effective for the multi-hop domain.

3.4. Cyclic Spiral Flow. Each spiral is one complete cycle: planner emits `<step>` → executor retrieves → state updates. Confidence increases monotonically with novel facts. Entropy decreases. Convergence occurs when confidence reaches threshold (0.85), entropy stalls for three consecutive spirals, or safety limits are hit (15 max hops, 4096 token budget).

3.5. LLM Backends. Three backends are implemented: MockBackend (canned responses, zero deps for testing), OllamaBackend (local CPU, no API key), and OpenAIBackend (production GPU cluster). Switching is a one-line config change.

§4. EXPERIMENTS

4.1. Setup. We evaluate on a multi-domain synthetic corpus (60 chunks, 10 domains) with 50 question-answer pairs. The model is qwen2.5:7b running on CPU via Ollama. We run two independent trials.

4.2. Metrics. We report: **Contains** (ground truth is a substring of prediction), **Token F1** (token-level overlap), **Exact Match** (strict equality), and **Spirals** (average cycles per question).

Metric	Run 1	Run 2	Avg
Contains	72%	70%	71%
Token F1	23%	22%	22.5%
Exact Match	0%	0%	0%
Avg Spirals	1.3	1.3	1.3
Time/QA (s)	324	336	330

Table 1. Benchmark results on 50 multi-domain QA (qwen2.5:7b CPU).

4.3. Results.

4.4. Analysis. Contains 70–72% with 1.3 average spirals validates the cyclic-spiral architecture. EM=0% is a format limitation: qwen2.5:7b outputs full sentences (e.g. “Jane Austen was born in 1775.”) against ground truth phrases (e.g. “1775”). This is not an architectural issue—Phase 2 with GPT-4o-mini targets EM 40%+. The spiral distribution confirms efficiency: 64% of questions solve in 0–1 spirals, 31% in 2, and only 5% in 3+.

§5. CONCLUSION

VORTEX demonstrates that a fact-free planner with fixed chained retrieval and entropic collapse is a viable architecture for multi-hop QA. Contains 72% on 7B CPU validates the approach. The system is offline-first, config-driven, and scales from laptop to cluster without code changes. Phase 2 will extend evaluation to the full HotpotQA benchmark and integrate semantic search via FAISS.

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CONTRIBUTION

dizel0110: conceptualization, implementation, benchmarking, paper writing.
Valentin Malykh: mentorship, architecture review, experimental design.

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